Invited Paper

High precision, medium flux rate CZT spectroscopy for coherent scatter imaging

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ABSTRACT

CZT detectors are primary candidates for many next-generation X-ray imaging systems. These detectors are typically operated in either a high precision, low flux spectroscopy mode or a low precision, high flux photon counting mode. We demonstrate a new detector configuration that enables operation in a high precision, medium flux spectroscopy mode, which opens the potential for a variety of new applications in medical imaging, non-destructive testing and baggage scanning. In particular, we describe the requirements of a coded aperture coherent scattering X-ray system that can perform fast imaging with accurate material discrimination.

Keywords: X-ray detectors, X-ray spectroscopy, X-ray diffraction

1. INTRODUCTION

Most digital radiation detectors are based on integrating charge of the X-ray photons emitted from the X-ray tube for each frame. Traditional systems use scintillator materials that convert X-ray photons into visible light and photo diode that in turn convert optical signals into electrical charge.[1] This technique is vulnerable to noise due to variations in the magnitude of the electric charge generated per X-ray photon and leads to a larger weighting of the higher energy photons and, consequently, reduced image contrast. X-ray photon counting detectors solve these problems by detecting directly the X-ray photons and assigning each a uniform weight. When photon counting is combined with energy discrimination, where the number of photons within a specific energy range are counted, one eliminates the noise associated with photon weighting and increases the information measurement capacity of the detector. Compared to energy integrating detectors, photon counting detectors offer a variety of potential advantages, including improved signal to noise ratio (SNR), reduction of beam hardening artifacts, reduced electronic noise, reduced dose, and improved tissue differentiation and/or material discrimination.

Over the last two decades, the II-VI semiconductor CdZnTe (CZT) has emerged as a material of choice for room temperature photon detection of hard X-rays and soft γ-rays. Due to improvements in crystal growth, the design of the detectors and the electronics used for reading out the detectors, CZT is now used in numerous applications including spectroscopy in astrophysics, industrial, medical imaging and security applications.[2,3] CZT-based detectors are typically operated in one of two modes: photon counting or spectroscopy. Photon counting mode offers modest energy resolution (ER, typically on the order of 10-20%, see Fig. 1) and can work at very high incident count rates (CR) of up to 100 Mcps/mm^2. In contrast, spectroscopy mode involves the precise determination of each photon's energy (with a precision of approximately 1%) but is limited to count rates of 1 kcps/mm^2 due to additional signal processing requirements. Nevertheless, there is a critical window between these two traditional modes in which the count rates are significantly higher than spectroscopy (*e.g.*, 10 -100 kcps/mm^2) and the required energy resolution is much better than required for photon counting (*e.g.*, 2-3%). It is precisely this medium count rate range that is of interest to many new emerging applications in non-destructive X-ray testing, baggage scanning and medical imaging.[4–7]

In this paper, we focus on the development of a new CZT-based detector that operates in this high precision medium flux regime for rapid coherent scatter X-ray tomography. While coherent scatter (*i.e.*, X-ray diffraction, XRD) has long been used for non-destructive examination of unknown objects due to its inherent molecular

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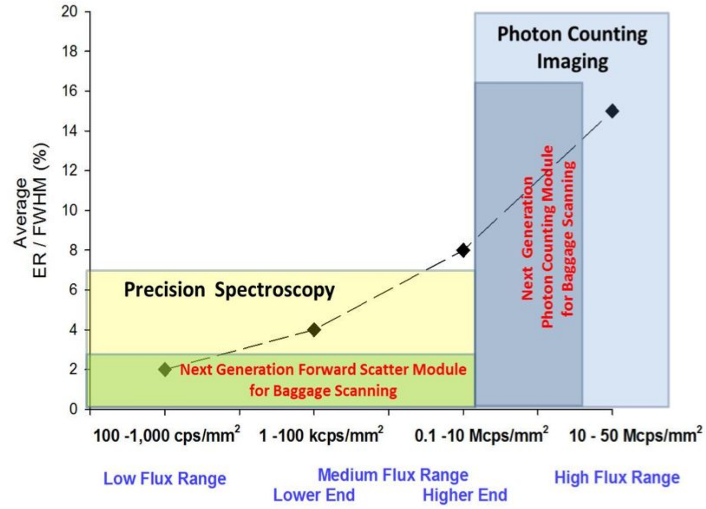

Figure 1. Schematic of the tradeoff between energy resolution and acceptable count rate in conventional CZT-based X-ray detectors. Spectroscopic detectors (left) provide good energy resolution at low count rates, whereas photon counting detectors (right) provide some energy resolution at high count rates.

specificity, it has traditionally been a non-imaging modality. Due to coherent scatter's potential for automatic anomaly detection in areas as diverse as cancer or explosives detection in situ, there has recently been much interest in performing coherent scatter tomography. Initial efforts of realizing XRD tomography were successful in producing images but typically required long scan times and technically complicated collimators and/or mechanical scanning apparatuses.[8,9] Recent advances in computational imaging and compressed sensing have enabled a new approach to XRD tomography through the use of coded apertures, which can increase the detectable scatter signal by several orders of magnitude.[6,10] To realize the potential for fast and accurate material identification with these systems, a new type of detector capable of both exquisite energy resolution and high count rates is required. Here we describe and characterize the first 2D spectroscopic X-ray detector optimized for coded aperture X-ray scatter imaging (see Fig. 2).

2. CZT DETECTOR OPTIMIZATION

For fast coherent scatter imaging systems, it is critical that the employed detector have both a small uncertainty in the energy and angle of the detected photons as well as the capacity to accept significant count rates. The angular resolution is largely determined by the spatial resolution of the system, measurement geometry, and the detector pixel size; in general, smaller pixels yield better angular resolution. The energy resolution is determined by the response of the detector material and associated signal processing chain (both in software and hardware). Taken together, the energy and angular resolution directly impact the ability to identify different materials, as they contribute to the uncertainty in the momentum transfer (or material lattice spacing).[6] Relatedly, the count rate (assuming sufficient quantum efficiency) determines the measurement time; a detector that can accept higher incident count rates can achieve a measurement of sufficient accuracy in a shorter time compared to one limited to lower count rates. The optimization of such a detector relies on the careful selection of CZT quality, pixel characteristics and layout, and signal processing components, as discussed below.

2.1 Flux range considerations

CZT detectors fabricated for low flux applications (less that 100 cps/mm^2) traditionally have a good electron mobility-lifetime product ($\mu\tau \sim 10^{-2}$ cm^2/V), but poor hole mobility-lifetime product ($\mu\tau \sim 10^{-5}$ cm^2/V). For that reason, most detectors in this class use single polarity readout schemes, where the main information about

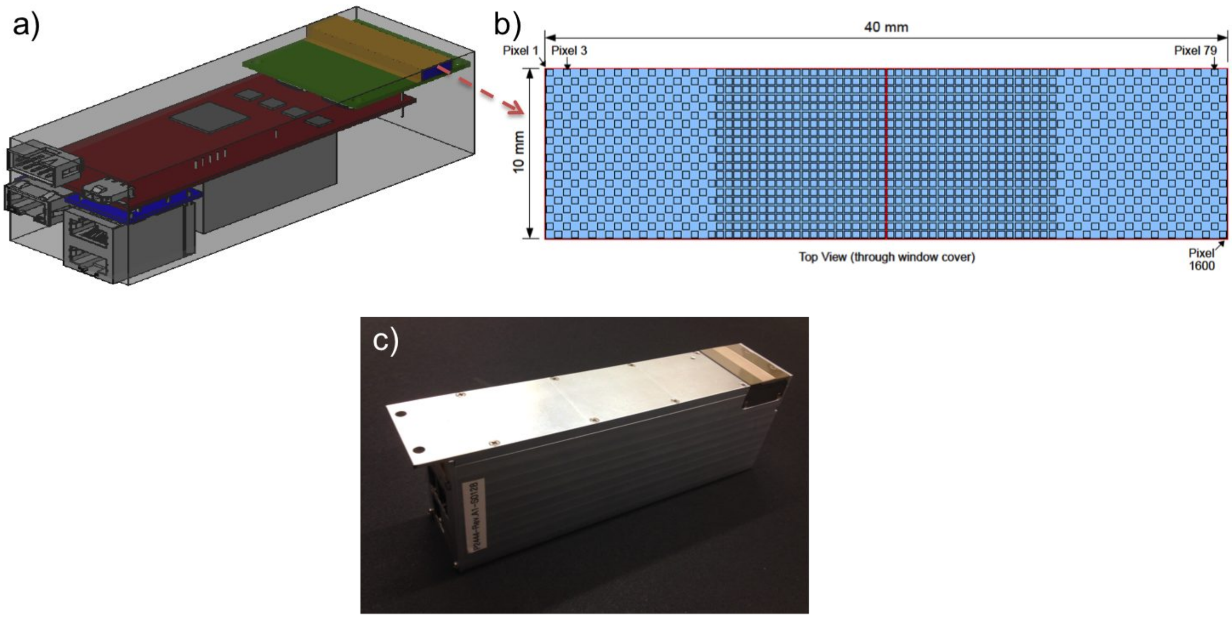

Figure 2. a) Three-dimensional representation of the CZT-based detector module. The yellow region is the active detector area. b) The pixel layout showing the 80 by 20 grid of 0.5 mm square pixels. Note that the pixels at the extreme left and right of the detector are arranged in a checkerboard pattern, whereas the pixels in the interior are arranged in a contiguous fashion. c) Photograph of the completed detector.

the energy of the detected radiation is inferred from the anode signals. Coplanar grid detectors (CPG), pixelated detectors, and Frisch grid detectors can overcome the severe hole trapping problem and greatly improve the energy spectra resolution of large-volume CZT detectors. Some state-of-the-art detector systems also utilize cathode signals in order to improve energy resolution (ER) using various depth of interaction (DOI) correction algorithms.[11] The continuous progress in CZT crystal growth and detector fabrication has lead to routinely realized energy resolutions of 1% with 10-15 mm thick pixelated detectors, with the best pixels achieving under 0.5% ER (which is close to the fundamental limit given by the Fano factor).

High flux operation (greater than 1 Mcps/mm^2) of CZT detectors brings additional challenges due to the polarization of the CZT. The origin of this effect can be traced back to high densities of crystal impurities and defects that, in turn, lead to severe charge-carrier trapping. Under high flux conditions, the trapped charge builds up inside the detector and effects its stability; in extreme conditions, this leads to complete collapse of the electric field and failure of the device. Recent advances in traveling heater method (THM) growth that require additional processing steps have enabled dramatic improvements of the hole mobility-lifetime product by an order of magnitude. As a result, high flux operation at count rates higher than 100 Mcps/mm^2 are routinely demonstrated. These improvements allow for the use of CZT in high flux environments, such as in computed tomography (CT). While this leads to significant advantages, such as reduced dose and improved image quality compared to scintillator-based detectors, these high flux detectors typically use semi-Ohmic contacts to avoid polarization effects which, in turn, results in higher dark currents and a degradation of the achievable energy resolution.

Coherent scatter imaging based on coded apertures resides somewhere between these two flux regimes. Assuming a nominal X-ray source voltage of 160 keV, a source current of between 1-10 mA (sufficient for scatter measurements on the time scale of a second or below), a coded aperture transmission of 50%, and a source-to-detector distance of 1 m, we typically find that the measured scatter signal is between 1-100 kcps/mm^2. While this exceeds the low flux regime by several orders of magnitude, we still require an energy resolution of only a few percent in order to maintain sufficient material discriminability. To meet these requirements, we targeted an energy resolution of 5% at 60 keV and a maximum count rate of 10 kcps/mm^2.

2.2 Pixel considerations

Given the specifications decided upon above, we next consider the thickness of the CZT material. Typical baggage scanning applications use a source voltage of 160 keV; in order to stop 95% of the incoming radiation, at least 3 mm of CZT is required. While choosing thicker CZT would increase the overall quantum efficiency, it can become bulky, costly, and potentially degrade the overall system energy resolution. We therefore consider a material thickness of between 2-5 mm.

For a given detector thickness, there is an optimum pixel size that maximizes energy resolution. If the pixel is too large, the small pixel effect (which confines the induced charge to a given anode) cannot occur and the energy resolution suffers. However, if the pixel is too small, the electron cloud resulting from diffusion process becomes comparable to (or even larger than) the pixel resulting in excessive charge sharing and charge loss in the inter-pixel gap. Numerical results indicate that the optimum pixel size for 2-5 mm thick CZT sample is somewhere between 0.5mm and 2.5 mm depending whether the charge shared events are corrected or not. We choose to use 0.5 mm pixels with a 3 mm thick sample of CZT to ensure good angular resolution of the overall imaging system.

2.3 Pixel layout

It is well-known that coherent scatter from liquids and polycrystalline powders give rise to azimuthally symmetric scatter rings; in contrast, crystalline materials with 'texture' (*i.e.*, a preferred orientation of the crystalline lattice) give rise to Bragg scatter that is concentrated at a particular angle for a given energy and lattice orientation. It is therefore necessary to employ 2D detectors to properly measure the coherent scatter signal for arbitrary objects. Relatedly, coherent scatter in the hard X-ray regime is usually limited to deflection angles of less than 10 degrees, which therefore suggests that the range of angles that one is required to measure is relatively low. To accommodate a fan beam geometry, which is common in baggage scanners, one therefore requires a detector that spans the length of the fan beam while also having an extent in the direction normal to the fan beam plane. To accommodate these constraints, we decided to create a 2-side buttable detector using the pixel layout shown in Fig. 2 b). The detector consists of 80 x 20 pixels composed of two, tiled CZT elements (each with dimensions of 20 mm by 10 mm) to create 40 mm by 10 mm almost continuous detection area with a very small gap between the two).

The ASIC is connected to the CZT via an interposer PCB board, which helps to provide thermal and structural isolation between the power dissipating ASIC and temperature sensitive CZT. At the same time, this extra element introduces additional stray capacitance, leading to a degradation of the system's energy resolution. Furthermore, it requires careful signal routing to avoid signal cross-talk. Due to differences in the ASIC footprint (20 by 20 mm centered on the detector) and the pixel layout (40 by 10 mm), the routing of the 1600 potential pixels became problematic. We therefore chose to connect only 1200 of the pixels, resulting in a checkerboard pattern for the pixels at the sides of the detector (as shown in Fig. 2). While this allows the remaining pixels to be read out, the extra routing complexity and path length results in poorer energy resolution for the pixels in the checkerboard region.

3. DETECTOR PERFORMANCE

Given the design choices and specifications described above, several prototype detectors were built (see Fig. 2). To characterize the detector, we use an extended, 4.95 mCi source of Am-241 (dominant emission line at 59.54 keV) placed approximately 100 mm from the detector. Figure 3 a) shows the recorded signal intensity (*i.e.*, total number of counts over all energies) in units of cps/mm^2. Note that the expected count rate is approximately 1500 cps/mm^2, indicating an overall detected quantum efficiency of approximately 30-50%. We generally find a uniform pixel response in the central region (except for some hot pixels) with reduced performance in the checkerboard regions, as expected.

We next consider the energy resolution across the detector. Figure 3 b) shows the full width at half maximum (FWHM) energy resolution (in units of keV) at each pixel throughout the detector. In the central region, the average energy resolution is 5% at 60 keV (2.98 keV FWHM) with a minimum energy resolution of 3.2% (1.89 keV FWHM). In the checkerboard region, the energy resolution ranges from 6-15% (roughly 4 to 10 keV FWHM).

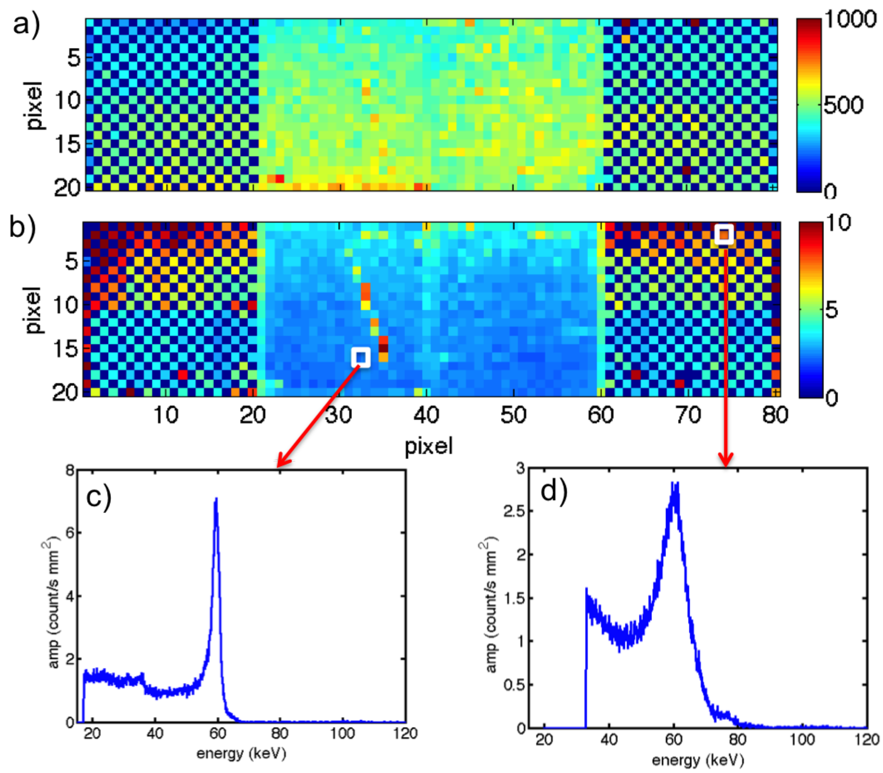

Figure 3. a) Distribution of pixel count rates across the detector in units of cps/mm^2. b) Energy resolution at each detector pixel at a count rate of approximately 500 cps/mm^2 in units of keV (FWHM) for a source at 60 keV. c) and d) show the recorded spectrum for Am-241 from pixels with good and poor energy resolution, respectively.

While energy resolution gives a first-order description of the spectroscopic response of the system, we show in Figs. 3 c) and d) examples of spectra from pixels with excellent and poor performance, respectively. For the pixel with low energy resolution, it can be seen that the spectrum falls off quickly above the photo peak, while residual counts at lower energies persist at approximately 10-20% of the photo peak amplitude. For the poorly performing pixel (Fig. 3 d), counts persist at a broader range of energies above and below the photo peak. Despite the anticipated poor performance at the ends of the detector, the energy resolution in the central portion of the detector is fairly uniform.

While the detector characterization was performed here at only 500 cps/mm^2, we observe uniform results up to approximately 10 kcps/mm^2, where this number refers to the incident count rate (rather than detected counts). We therefore achieved the target specifications of less than 5% energy resolution at count rates approaching 10 kcps/mm^2 in the central 800 pixels of the detector.

4. CONCLUSIONS

Advances in CZT technology make possible pixellated arrays of energy-sensitive detectors at room temperature and for hard X-rays. Such detectors are critical for real-time coherent scatter imaging in applications such as baggage inspection and medical imaging, but the introduction of new coded aperture scatter imaging architectures require detectors that can perform high-precision spectroscopy at medium flux rates. To accommodate this new approach to coherent scatter imaging, we developed a detector capable of yielding excellent angular and energy resolution (less than 5%) as well as being able to operate at up to 10 kps/mm^2. While this detector will be an invaluable tool in next generation coherent scatter imaging systems, there are a number of future improvements that can be made to the detector. Examples of such revisions include implementing charge sharing corrections to reduce the amplitude of the spectral tails, revising the pixel layout to better match the ASIC footprint and/or increasing the pixel pitch to mitigate routing issues and improve uniformity, and avoiding the necessity of the checkerboard pattern to allow for tiling of the detectors into a contiguous array. In addition, we plan to study directly coherent X-ray scatter with these detectors to better understand the the detector's capabilities.

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